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Regio- and Stereoselective Synthesis of Rotationally Hindered C_{12}-Naphthylated Tribenzo[*a,c,j*]xanthenes through Catalytic Tricyclization of Yne-Allenones

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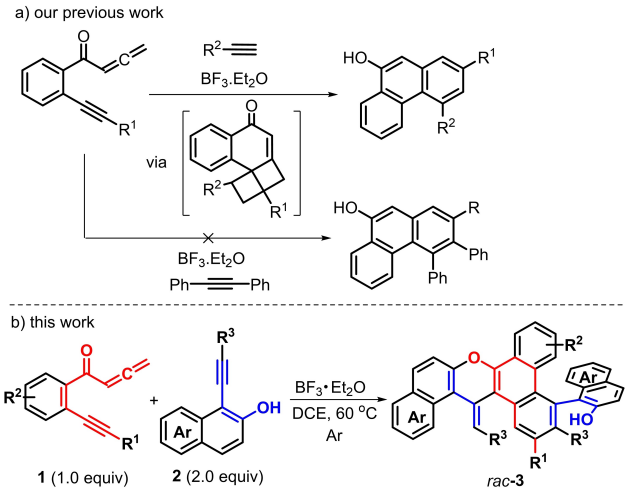
Abstract: A Lewis acid-catalyzed tricyclization of yne-allenones with 1-alkynylnaphthalen-2-ols has been developed, and used to produce a wide range of rotationally hindered tribenzo[*a,c,j*]xanthenes with generally good yields and stereoselectivity. This process involves a complex [2+2] cycloaddition relay and [4+2] cyclization cascade, enabling scission/recombination of C–C triple bonds to access unprecedented hexaheterocyclic architectures. The reaction also features good compatibility with different types of substituents, triple annulation cascade and the efficient control of regio- and stereochemistry.

Keywords: Alkynes; Oxygen heterocycles; Regioselectivity; Annulation; Multicomponent reactions

Functional oxa-heterocycles, due to their prevalence in substantial auxiliaries,[1] building blocks,[2] biological substances[3] and clinical drugs,[4] are always in great demand in organic synthesis, chemical biology, and material science as well as medicinal chemistry.[5] Consequently, the development of efficient and practical synthetic methodologies for the assembly of heterocycles with challenging oxa-skeletons such as xanthenes and stereochemistry becomes one of important targets.[6] Among the enormous methods, multiple cyclization strategy is an attractive approach and has been widely applied for the generation of structurally complex oxygen-containing molecules because of their inherent characteristics of annulation cascade, step- and atom-economy and sustainability as well as convergence.[7]

Cycloaddition reactions have evolved into effective technique to assemble cyclic rings in the organic community.[8] Over the past decades, various reaction types, including [2+1],[9] [2+2],[10] [3+2],[11] [4+2],[12] and [4+4][13] cycloaddition and so on,[14] have been well-developed by means of thermal, organocatalytic, photoinduced or/and transition metal-catalyzed strategies. In contrast, standing for a new mode of cycloaddition reaction, double [2+2] cycloaddition relay remains elusive and constitutes a much more challenging task, because of their cycloaddition partners must match their frontier molecular orbitals. Very recently, we reported a first catalytic double [2+2] cycloaddition relay between yne-allenones and unactivated alkenes or alkynes, leading to the formation of functionalized phenanthren-9-ols through the C–C triple bond breaking/rearranging (Scheme 1a).[15] However, the reaction has suffered from a major restriction that the electron-neutral internal arylated alkynes such as 1,2-diphenylethyne remained unreactive in this transformation. We analyzed that the frontier molecular orbital of such internal alkyne did not match that of *in-situ*-generated activated alkene from yne-allenones, thus requiring the higher electron density of the internal alkynes. On the other hand, 1-alkynyl naphthalen-2-ols serve as electron-rich internal alkynes showed a high reactivity profile, enabling their intermolecular [2+2] cycloaddition reactions with electron-efficient alkenes to construct *oxa*-heterocycles.[15] Inspired by these successful trans-

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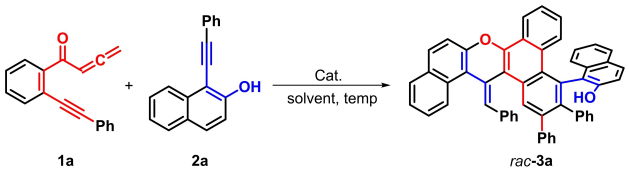


Scheme 1. Tricyclization of yne-allenones toward tribenzo [*a,c,j*]xanthenes.

formations, we focused on the feasibility of double [2 + 2] cycloaddition relay between 1- yne-allenones **1** as electron-deficient olefins and 1-alkynylnaphthalen-2-ols **2** as electron-rich internal alkynes. While carrying out this project, unprecedented rotationally hindered C_{12}-naphthylated tribenzo[*a,c,j*]xanthenes **3** with the satisfactory control of regio- and stereochemistry were obtained through a $BF_3 \cdot Et_2O$-catalyzed reaction of **1** with **2** in 1:2 mole ratio (Scheme 1b). It can be noted that the current protocol represents a first Lewis acid-catalyzed tricyclization cascade involving double [2 + 2] cycloaddition relay for the regioselective synthesis of the hitherto unreported hexacyclic tribenzo[*a,c,j*] xanthenes through catalytic C–C triple-bond cleavage of yne-allenones. This reaction features a good substrate scope and high atom utilization, providing an extensively attractive entry to assemble polycyclic architectures. Herein, we elaborate this three-component tricycliczation reaction starting from yne-allenones and 1-alkynylnaphthalen-2-ols, enabling the direct formation of rotationally hindered hexacyclic molecular scaffolds.

Our investigation commenced with the reaction of yne-allenone **1a** with 1-alkynyl naphthalen-2-ol **2a** in a 1:2 mol ratio in the presence of $BF_3 \cdot Et_2O$ (10 mol%) at 50 °C in 1,2-dichloroethane (DCE) under argon conditions. The reaction worked smoothly, affording the unexpected hexacyclic product **3a** in 75% yield with 4:1 *Z*/*E* ratio (Table 1, entry 1). The following screening of other solvents including dichloromethane (DCM), MeCN, 1,4-dioxane and EtOH indicated that all these attempts did not improve the efficiency of this reaction, resulted in a remarkably reduced yield of **3a** (entries 2–5). Next, several other Lewis acid metal catalysts, such as $Sc(OTf)_3$, AgOTf, $Sn(OTf)_2$, $FeCl_3$ and $Y(OTf)_3$, were also examined. The results showed

Table 1. Optimization of reaction conditions for **3a**.[a]



Entry	Cat. (mol %)	Solvent	*t* (°C)	Yield (%)[b]
1	$BF_3 \cdot Et_2O$ (10)	DCE	50	75
2	$BF_3 \cdot Et_2O$ (10)	DCM	50	65
3	$BF_3 \cdot Et_2O$ (10)	MeCN	50	43
4	$BF_3 \cdot Et_2O$ (10)	1,4-dioxane	50	25
5	$BF_3 \cdot Et_2O$ (10)	EtOH	50	N.R.
6	AgOTf (10)	DCE	50	67
7	$Sc(OTf)_3$ (10)	DCE	50	39
8	$Sn(OTf)_2$ (10)	DCE	50	46
9	$FeCl_3$ (10)	DCE	50	35
10	$Y(OTf)_3$ (10)	DCE	50	55
11	$BF_3 \cdot Et_2O$ (10)	DCE	60	78
12	$BF_3 \cdot Et_2O$ (10)	DCE	40	64
13	$BF_3 \cdot Et_2O$ (20)	DCE	60	72
14[c]	$BF_3 \cdot Et_2O$ (10)	DCE	60	83
15[c]	$BF_3 \cdot Et_2O$ (10)	dry DCE	60	87
16[c]	none	dry DCE	60	N.R.

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.4 mmol), $BF_3 \cdot Et_2O$ (10 mol%), solvent (2.5 mL), under argon conditions, 12 h.
[b] Isolated yield based on substrate **1a**. N.R. = no reaction.
[c] **1a** and **2a** in 1:2.2 mole ratio

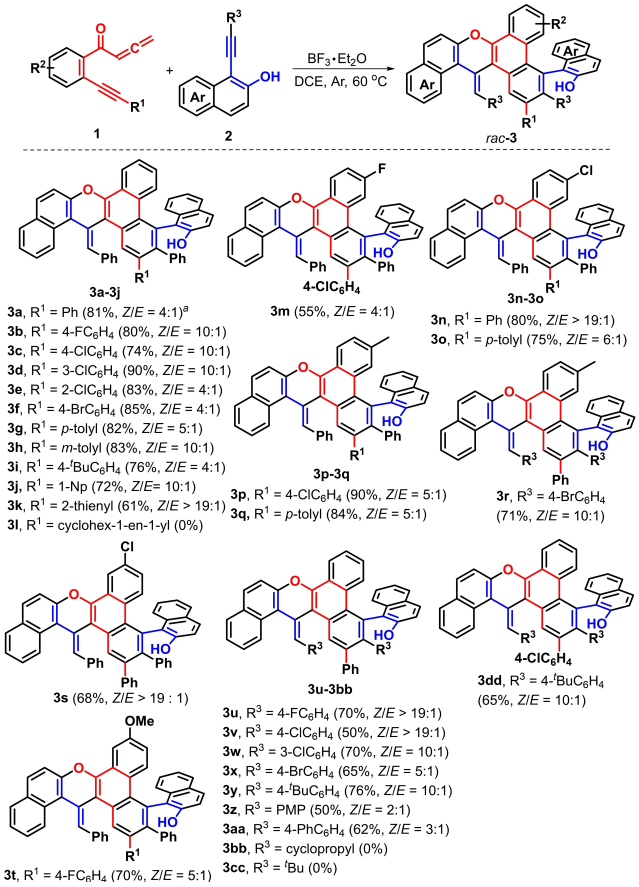
that these five catalysts could promote the formation of **3a**, but all demonstrated lower catalytic capabilities and provided unsatisfactory yields in comparison with $BF_3 \cdot Et_2O$ (entries 6–10 vs entry 1). It is found that the reaction temperature could profoundly impact on the yield of **3a**. The reaction worked more efficiently, giving a slightly higher yield of product **3a** (78%) when the temperature was elevated to 60 °C (entry 11) whereas a decrease in the temperature from 50 to 40 °C led to the lower conversion, delivering a profoundly dropped yield (64%) as the starting materials were not completely consumed, even after prolonging the reaction time (entry 12). Moreover, an increased loading of $BF_3 \cdot Et_2O$ failed to facilitate this transformation (entry 13). Fine-tuning the mole ratio of **1a** and **2a** to 1:2.2 is beneficial for this tricyclization process, giving 83% yield (entry 14). With dry DCE being used, a higher yield was gained (87%, entry 15). For this reason, the reaction was routinely carried out in dry DCE. Without $BF_3 \cdot Et_2O$, the reaction did not proceed with unreacted **1a** and **2a** in 100% recovery (entry 16).

Having demonstrated the viability of engaging the yne-allenone **1a** in the tricyclization reaction (Table 1, entry 15), we then systematically evaluated the generation and limitation of this protocol. The results

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illustrated in Scheme 2. Initially, the effects of changing of the electronic properties and positions of substituents in the arylalkynyl moiety (R^1) was investigated by carrying out the reaction of various yne-allenone **1** with **2a**, and all of them could participate in the current catalytic tricyclization with high efficiency. A variety of substituents, such as fluoro (**1b**), chloro (**1c**–**1e**), bromo (**1f**), methyl (**1g**–**1h**) and *tert*-butyl (**1i**), residing in different positions (*para, meta* or *ortho*) of the arene ring were suitable for this catalytic system, delivering the desired hexacyclic (*Z*)-products **3b**–**3i** in good to excellent yields with 4:1 to 10:1 *Z*/*E* ratio (Scheme 2). Of these groups, sterically encumbered *o*-chlorophenyl counterpart (**1e**) demonstrated good compatibility, indicating that the increased steric congestion had no influence on the course of the reaction. The 1-naphthalenyl- (1-Np, **1j**) and 2-thienyl-containing (**1k**) substrates were employed to prove the compatibility of this tricyclization, furnishing the corresponding products **3j** and **3k** with synthetically useful yields and high stereoselectivity. We also prepared yne-allenone **1l** with a cyclohex-1-en-1-yl group that failed to give **3l** in tractable amount. Subsequently, the electronic nature of substituents (R^2) on the internal arene ring of substrates **1** was briefly examined. Several representative substituents such as fluoro (**1m**), chrolo (**1n**–**1o** and **1s**), methyl (**1p**–**1r**) and methoxy (**1t**) were independently introduced into the C4 or C5 position of the internal arene ring and then these substrates (**1m**–**1t**) were reacted with **2a** under the standard conditions. To our pleasure, these groups could be accommodated, producing the corresponding products **3m**–**3t** with acceptable yields and stereoselectivity. Next, the generality of this protocol was further assessed by investigating a brief survey of the possible variation in the structures of 1-alkynylnaphthalen-2-ols in combination with **1a**. With a standard operation (Table 1, entry 15), 1-alkynyl naphthalen-2-ols with diverse electronic nature and different substituents at different positions of the phenyl ring relative to the arylalkynyl moiety (R^3) were probed. Both electron-withdrawing (e.g., fluoro, **2b**; chloro, **2c**–**2d**; bromo, **2e**) and electron-donating (e.g., *tert*-butyl, **2f**; methoxy (PMP = p-methoxyphenyl), **2g**; phenyl, **2h**) groups could generate the corresponding products **3u**–**3aa** with comparable efficiency. Unluckily, exchanging the aryl fragment in the alkynyl moiety for an alkyl group such as cyclopropyl or *tert*-butyl did not give access to the desired products **3bb** and **3cc**, suggesting that the aryl group stabilized 1-alkynylnaphthalen-2-ols prove to be good precursors. Moreover, upon treatment of **1c** with **2f** enabled a similar tricyclization process to afford product **3dd** with 65% yield and 10:1 *Z*/*E* ratio. Of note, this transformation provides an efficient and practical entry for the rapid construction of a xanthene-containing hexacyclic architectures bearing a rotationally hindered chiral axis in one step under the mild conditions. The stereo-structure of **3b** was confirmed by single crystal X-ray diffraction (Figure 1, CCDC 2054923),[16] and their derivatives were assigned by analogy (see Supporting Information).

To demonstrate the potential application of this method, an amplification reaction was performed under



Scheme 2. Substrate scope for the synthesis of **3**.

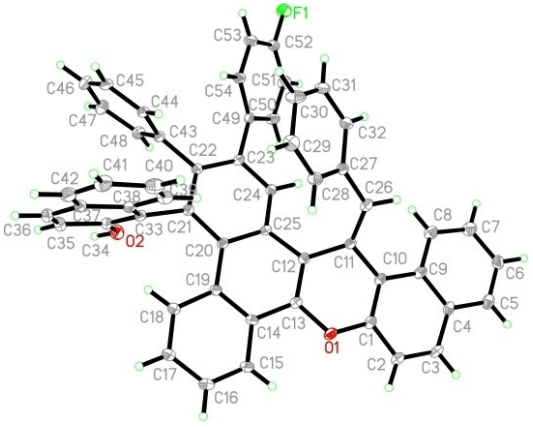
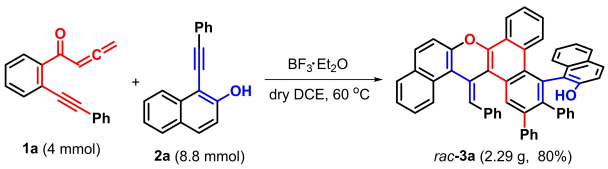


Figure 1. X-Ray structure of product **3b**.

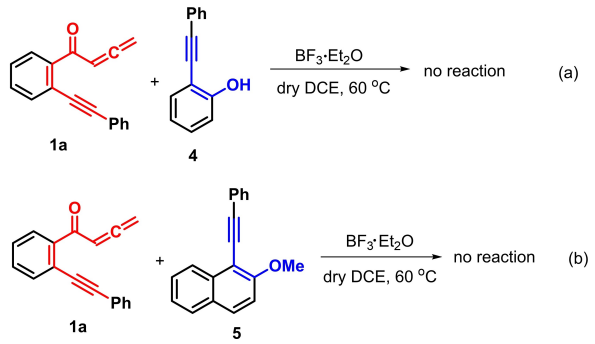
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the standard conditions. We were delighted to find that product **3 a** was obtained in 80% yield on a 4 mmol scale (Scheme 3).

To gain mechanistic insight into this tricyclization, several control experiments were conducted. The reaction of **1 a** with 2-(phenylethynyl)phenol **4** did not work under the standard conditions and starting materials were recovered (Scheme 4a). Similarly, *O*-protected 1-alkynylnaphthalen-2-ol **5** did not react with **1 a** as well (Scheme 4b). These observations suggested that naphthol ring linked by the alkynyl unit is crucial for the current tricyclization.



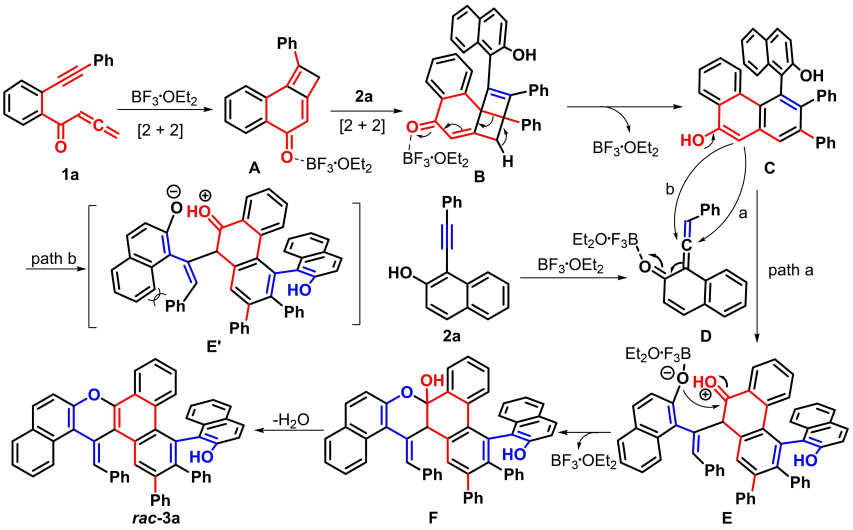
Scheme 3. Scale-up transformation of **3 a**.



Scheme 4. Preliminary mechanistic investigation.

By combining our experimental results with previous reports,[15,17] a reasonable pathway for forming products **3** was proposed in Scheme 5. Yne-allenone **1 a** first undergoes an intramolecular [2+2] cycloaddition to offer adduct intermediate **A**,[15] followed by $BF_3 \cdot Et_2O$-catalyzed intermolecular [2+ 2] cycloaddition with 1-alkynylnaphthalen-2-ol **2 a** to yield intermediate **B**. Then, ring-opening of four-membered carbocycle of intermediate **B**, because of its ring tension, easily occurs under the catalytic condition, giving six-membered ring intermediate **C**. The *in-situ* formed vinylidene *ortho*-quinone methide (VQM) intermediate **D**[18] from 1-alkynylnaphthalen-2-ol **2 a** in the presence of $BF_3 \cdot Et_2O$ is intercepted by intermediate **C** to give intermediate **E** (path a), which is converted into the final hexacyclic (*Z*)-isomer **3 a** as the major stereoisomers through a catalytic *oxa*-cyclization and dehydration process. The stereoselectivity should be attributed the steric hindrance. The *in-situ* generated intermediate **E'** is converted into (*E*)-stereoisomer **3 a** as the minor stereoisomers due to its greater steric hindrance (path b).

In summary, we have developed a novel Lewis acid-catalyzed tricyclization cascade involving the [2+ 2] cycloaddition relay, enabling the scission/recombination of the C–C triple bonds to stereoselectively generate hitherto unreported hexacyclic tribenzo[*a*,*c*,*j*] xanthenes bearing a rotationally hindered chiral axis with generally good yields. The reaction progress is considered to proceed through a double [2+2] cycloaddition relay, C–C triple bond cleavage and *oxa*-cyclization sequence. The three-component tricyclization favors the efficient control of regio- and stereochemistry, good compatibility with different types of substituents, high atom utilization as well as triple annulation cascade. Further investigation and applica-



Scheme 5. Plausible reaction pathway.

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tion of this tricyclization strategy is underway in our laboratory.

Experimental Section

General Procedure for the Synthesis of Compounds 3 aa

To a 10 mL Schlenk tube under Ar conditions, 1-(2-(phenylethynyl)phenyl)buta-2,3-dien-1-one (**1 a**, 0.2 mmol, 48.8 mg, 1 equiv.), 1-(phenylethynyl)naphthalen-2-ol (**2 a**, 0.44 mmol, 107.36 mg, 2.2 equiv.), $BF_3 \cdot Et_2O$ (10 mol%, 2.82 mg), and dry 1,2-dichloroethane (2.5 mL) were successively added. Then, the reaction tube was stirred at 60 °C for 12 h until complete consumption of **1 a**, as monitored by TLC analysis. After the reaction was completed, the reaction mixture was concentrated in vacuum and the resulting residue was purified by column chromatography on silica gel (eluent, petroleum ether/ethyl acetate = 5:1) to afford the desired product **3 a** as light-yellow solid

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References

[1] a) D. Alessandro, *Org. Biomol. Chem.* **2010**, *8*, 3366–3385; b) K. Dorota, C. Jacek, J. Aleksandra, D. Jozef, *Phosphorus Sulfur Silicon Relat. Elem.* **2017**, *192*, 585–588; c) P.-F. Stephen, *Curr. Org. Chem.* **2004**, *8*, 1289–1337.

[2] a) R. Pratap, V.-J. Ram, *Tetrahedron Lett.* **2017**, *73*, 2529–2590; b) V.-F. Batista, *Molecules* **2020**, *25*, 3791–3811.

[3] a) M.-E. Sousa, M.-M. Pinto, *Curr. Med. Chem.* **2005**, *12*, 2447–2479; b) G. Arefg, N. Zahras, N. Seyedmr, *Curr. Org. Synth.*, **2019**, *16*, 1071–1077.

[4] a) D. Michaeld, S. Davidt, A. Bentonj, N. Jont, *J. Med. Chem.* **2018**, *61*, 10996–11020; b) S. Kinga, M. Andrzej, L. Tadeusz, *Mini-Rev. Med. Chem.* **2013**, *13*, 335–352.

[5] a) I. Haiduc, *J. Coord. Chem.* **2020**, *73*, 2117–2170; b) R.-S. Keri, B.-S. Sasidhar, M.-J. Bhari, M. Amélia-Santos, *Eur. J. Med. Chem.* **2015**, *100*, 257–256; c) H. Zhang, X.-C. Liu, Y.-X. Gong, T.-Z. Yu, Y.-L. Zhao, *Dyes Pigm.* **2021**, *185*, 108969–108975; d) M. Miguel, F. Durães, E. Sousa, *Eur. J. Med. Chem.* **2021**, *210*, 113085–113114.

[6] a) K. Takaishi, S. Hinoide, T. Matsumoto, E. Tadashi, *J. Am. Chem. Soc.* **2019**, *141*, 11852–11857; b) T. Kamei, M. Uryu, T. Shimada, *Org. Lett.* **2017**, *19*, 2714–2717; c) Z.-L. Tao, K.-A. Robb, K. Zhao, S.-E. Denmark, *J. Am. Chem. Soc.* **2018**, *140*, 3569–3573; d) S. Saha, C. Schneider, *Org. Lett.* **2015**, *17*, 648–651; e) X.-B. Xu, X.-L. Xu, H.-F. Li, X. Xie, Y.-Z. Li, *Org. Lett.* **2010**, *12*, 100–103; f) L.-M. Wysocki, J.-B. Grimm, A.-N. Tkachuk, T.-A. Brown, E. Betzig, L.-D. Lavis, *Angew. Chem. Int. Ed.* **2011**, *50*, 11206–11209; g) N. Kaur, N. Ahlawat, P. Grewal, P. Bhardwaj, Y. Verma, *Curr. Org. Chem.* **2019**, *23*, 2822–2847; h) A. Chaudhary, J.-M. Khurana, *Curr. Org. Synth.* **2018**, *15*, 341–369.

[7] a) S.-S. Zhu, J.-N. Zhou, Q.-L. Wu, W.-J. Hao, S.-J. Tu, B. Jiang, *Org. Chem. Front.* **2020**, *7*, 2975–2980; b) G. Mehta, G. Panda, *Chem. Commun.* **1997**, 2081–2082; c) M. Taguchi, Y. Tokimizu, S. Oishi, N. Fujii, H. Ohno, *Org. Lett.* **2015**, *17*, 6250–6253; d) B.-B. Feng, J.-Q. Liu, X.-S. Wang, *J. Org. Chem.* **2017**, *82*, 1817–1822; e) B. Bennacer, M. Fujiwara, S.-Y. Lee, I. Ojima, *J. Am. Chem. Soc.* **2005**, *127*, 17756–17767; f) F. Bartels, Y. J. Hong, D. Ueda, M. Weber, T. Sato, D. J. Tantillo, M. Christmann, *Chem. Sci.* **2017**, *8*, 8285–8290; g) A. Gupta, B. L. Flynn, *Org. Lett.* **2017**, *19*, 1939–1941.

[8] a) J.-H. Wang, S.-A. Blaszczyk, X.-X. Li, W.-P. Tang, *Chem. Rev.* **2021**, *121*, 110–139; b) K.-O. Marichev, M.-P. Doyle, *Org. Biomol. Chem.* **2019**, *17*, 4183–4195.

[9] a) J. Barluenga, H. Fanlo, J. Florez, *Angew. Chem. Int. Ed.* **2007**, *46*, 4136–4140; b) Y. Lou, T.-P. Remarchuk, E. J. Corey, *J. Am. Chem. Soc.* **2005**, *127*, 14223–14230; c) N. Al-Bataineh, K.-N. Houk, M. Brewer, X. Hong, *J. Org. Chem.* **2017**, *82*, 4001–4005.

[10] a) E. Canales, E.-J. Corey, *J. Am. Chem. Soc.* **2007**, *129*, 12686–12687; b) M.-V. Joannou, J.-M. Hoyt, P.-J. Chirik, *J. Am. Chem. Soc.* **2020**, *142*, 5314–5330; c) K. Takasu, M. Ueno, K. Inanaga, M. Ihara, *J. Org. Chem.* **2004**, *69*, 517–521; d) B. Alcaide, P. Almendros, C. Aragoncillo, *Chem. Soc. Rev.* **2010**, *39*,783–816.

[11] a) Q.-H. Zha, Y.-K. Wu, *J. Org. Chem.* **2020**, *85*, 14121–14138; b) S.-S. Zhang, D.-C. Wang, M.-S. Xie, G.-R. Qu, H.-M. Guo, *Org. Lett.* **2018**, *20*, 8026–8029; c) B.-M. Trost, A.-H. Shinde, Y.-L. Wang, Z.-J. Zuo, C. Min, *ACS Catal.* **2020**, *10*, 1969–1975; d) A.-S. Kumari, K.-K. Swamy, *J. Org. Chem.* **2015**, *80*, 8, 4084–4096.

[12] a) T.-T. He, L. Peng, S. Li, F.-L. Hu, C.-D. Xie, S.-L. Huang, S.-Q. Jia, W.-L. Qin, H.-H. Yan, *Org. Lett.* **2020**, *22*, 6966–6971; b) C.-S. Sumaria, Y.-E. Türkmen, V.-H. Rawal, *Org. Lett.* **2014**, *16*, 3236–3239.

[13] D. Bailey, V.-E. Williams, *J. Org. Chem.* **2006**, *71*, 5778–5780.

[14] a) L. Wang, S. Li, M. Blumel, A.-R. Philipps, A. Wang, R. Puttreddy, K. Rissanen, D. Enders, *Angew. Chem. Int. Ed.* **2016**, *55*, 11110–11114; b) V. Srinivas, K.-V. Sajnaa, K.-K. Swamy, *Chem. Commun.* **2011**, *47*, 5629–5631; c) K.-V. Sajna, R. Kotikalapudi, M. Chakravarty, N.-B. Kumar, K.-K. Swamy, *J. Org. Chem.* **2011**, *76*, 3, 920–938; d) A.-S. Kumari, A.-S. Reddya, K.-K. Swamy, *Org. Biomol. Chem.* **2016**, *14*, 6651–6671; e) E. Haraburda, Ò. Torres, T. Parella, M. Solà, P.-Q. Anna, *Chem. Eur. J.* **2014**, *20*, 5034–5045; f) H. Ewelina, L. Agustí, R. Anna, P.-Q. Anna, *Org. Lett.* **2015**, *17*, 2882–2885; g) L. Dirk, A. Alexandra, F. Martin, R. Sebastian, D. Gerald, *Curr. Org. Chem.* **2019**, *23*, 3064–3067.

[15] H. Li, W.-J. Hao, M. Wang, X. Qin, S.-J. Tu, P. Zhou, G. Li, B. Jiang, *Org. Lett.* **2018**, *20*, 4362–4366.

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[16] CCDC 2054923 (**3b**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.

[17] a) J.-Y. Wang, P. Zhou, G. Li, W.-J. Hao, S.-J. Tu, B. Jiang, *Org. Lett.* **2017**, *19*, 6682–6685; b) F. Liu, J.-Y. Wang, P. Zhou, G. Li, W.-J. Hao, S.-J. Tu, B. Jiang, *Angew. Chem. Int. Ed.* **2017**, *56*, 15570–15574; c) C.-L. Ji, W.-J. Hao, J. Zhang, F.-Z. Geng, T. Xu, S.-J. Tu, B. Jiang, *Org. Lett.* **2019**, *21*, 6494–6498; d) K. Chen, T. Xu, J. Liang, M. Zhou, J. Zhang, W.-J. Hao, J.-Y. Wang, S.-J. Tu, B. Jiang, *Chem. Commun.* **2019**, *21*, 9784–9789; e) T. Xu, K. Chen, H.-Y. Zhu, W.-J. Hao, S.-J. Tu, B. Jiang, *Org. Lett.* **2020**, *22*, 2414–2418; f) T. Xu, N. Lin, W.-J. Hao, J. Zhang, M.-F. Li, S.-J. Tu, B. Jiang, *Chem. Commun.* **2020**, *56*, 11406–11409; g) T. Feng, Y. He, X.-Y. Zhang, X.-S. Fan, *Adv. Synth. Catal.* **2019**, *361*, 1271–1276; h) Y. He, T. Feng, X.-S. Fan, *Org. Lett.* **2019**, *21*, 3918–3922.

[18] For selected examples, see: a) L. Peng, D. Xu, X. Yang, J. Tang, X. Feng, S.-L. Zhang, H. Yan, *Angew. Chem. Int. Ed.* **2019**, *58*, 216–220; b) Y. Tan, Z. Zhao, Z. Chen, S. Huang, S. Jia, L. Peng, D. Xu, W. Qin, H. Yan, *Org. Lett.* **2020**, *22*, 4461–4466; c) P. Tian, C.-Q. Wang, S.-H. Cai, S. Song, L. Ye, *J. Am. Chem. Soc.* **2016**, *138*, 15869–15872; d) R. Huang, X. Chang, J. Li, C.-J. Wang, *J. Am. Chem. Soc.* **2016**, *138*, 3998–4001; e) L. Zhou, Z. Li, Y. Zou, Q. Wang, I.-A. Sanhueza, F. Schoenebeck, A. Goeke, *J. Am. Chem. Soc.* **2012**, *134*, 20009–20012.

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